

Lecture 5

Characters, the canonical decomposition, and projectors

Objectives. The trace compresses each representation matrix to a single number, and these numbers — the characters — turn out to retain everything that matters: they detect equivalence, count multiplicities, and test irreducibility, all through one inner product. We develop the two orthogonality relations, construct character tables for C_n , S_3 , D_4 and C_{3v} by a systematic algorithm, and convert the theory into operators: the projectors that carve any representation into its symmetry types, the computational workhorse of the applications in Lectures 6 and 7.

5.1 Characters and class functions

Definition 5.1 (Character). The character of a representation (ρ, V) of G is the function

$$\chi_\rho: G \rightarrow \mathbb{C}, \quad \chi_\rho(g) = \text{tr } \rho(g) = \text{tr } D(g),$$

the trace being the same in every basis. The characters of the irreducible representations $\rho^{(i)}$ are written χ_i and called *irreducible characters*. Note $\chi_\rho(e) = \text{tr id}_V = \dim V$: the character remembers the dimension.

Proposition 5.2 (Basic properties). Let χ be the character of (ρ, V) , with G finite. Then:

- (i) equivalent representations have equal characters;
- (ii) $\chi(hgh^{-1}) = \chi(g)$ for all g, h : characters are constant on conjugacy classes;
- (iii) $\chi(g^{-1}) = \overline{\chi(g)}$;
- (iv) characters add on direct sums: $\chi_{\rho \oplus \sigma} = \chi_\rho + \chi_\sigma$;
- (v) the dual representation has the conjugate character: $\chi_{\rho^*}(g) = \chi_\rho(g^{-1}) = \overline{\chi_\rho(g)}$.

Proof. (i) and (ii) are the invariance of the trace under conjugation: $\text{tr}(SDS^{-1}) = \text{tr } D$ and $\text{tr}(D(h)D(g)D(h)^{-1}) = \text{tr } D(g)$. (iii) The element g has finite order m , so $\rho(g)^m = \text{id}$; the minimal polynomial of $\rho(g)$ divides $x^m - 1$, which has distinct roots, so $\rho(g)$ is diagonalisable with eigenvalues $\lambda_1, \dots, \lambda_n$ that are m -th roots of unity. Then $\chi(g) = \sum_a \lambda_a$, while $\rho(g^{-1}) = \rho(g)^{-1}$ has eigenvalues $\lambda_a^{-1} = \overline{\lambda_a}$ (unit modulus), whence $\chi(g^{-1}) = \overline{\chi(g)}$. (iv) In a basis adapted to the direct sum the matrices are block diagonal, and the trace adds over blocks. (v) From Lecture 3, $D^*(g) = D(g^{-1})^T$, and transposition does not change the trace; then apply (iii). ■

Property (ii) places the characters in a distinguished finite-dimensional space:

Definition 5.3 (Class functions). A class function on G is a function $G \rightarrow \mathbb{C}$ constant on conjugacy classes. Class functions form a subspace $\text{CF}(G)$ of the space of all functions on G ,

of dimension equal to the number of conjugacy classes (one free value per class), which we endow with the inner product of Lecture 4,

$$\langle \varphi, \psi \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{\varphi(g)} \psi(g) = \frac{1}{|G|} \sum_{\text{classes } C} |C| \overline{\varphi(g_C)} \psi(g_C),$$

the second form (with g_C any representative of C) being how one computes in practice.

One more property of characters costs a line now and pays handsomely in Lecture 6:

Lemma 5.4 (Fixed-point count). *The character of a permutation representation (Lecture 3) counts fixed points:*

$$\chi_{\text{perm}}(g) = \#\{x \in X : g \cdot x = x\}.$$

Proof. The diagonal entries of the permutation matrix are $D_{xx}(g) = \delta_{x, g \cdot x}$: a 1 for each fixed point. ■

(In Lecture 6 this becomes the “atoms left in place” rule for the vibrational analysis of molecules.) For the regular representation, $X = G$ acted on by left translation, $g \cdot h = gh$ has a fixed point only for $g = e$:

$$\chi_{\text{reg}}(e) = |G|, \quad \chi_{\text{reg}}(g) = 0 \quad (g \neq e). \quad (5.1)$$

5.2 The orthogonality relations

Theorem 5.5 (First orthogonality relation). *The irreducible characters are orthonormal:*

$$\langle \chi_i, \chi_j \rangle_G = \delta_{ij}.$$

Since the number of irreducible representations equals the number of conjugacy classes (Lecture 4), the irreducible characters form an orthonormal basis of the space of class functions.

Proof. Take unitary matrices and sum the great orthogonality theorem of Lecture 4 over the diagonal:

$$\langle \chi_i, \chi_j \rangle_G = \sum_{\mu=1}^{n_i} \sum_{\alpha=1}^{n_j} \left\langle D_{\mu\mu}^{(i)}, D_{\alpha\alpha}^{(j)} \right\rangle_G = \sum_{\mu, \alpha} \frac{1}{n_i} \delta_{ij} \delta_{\mu\alpha} \delta_{\mu\alpha} = \delta_{ij} \frac{n_i}{n_i} = \delta_{ij}.$$

The characters are thus k orthonormal — hence linearly independent — vectors in the k -dimensional space $\text{CF}(G)$, where k is the number of classes: a basis. ■

The same information, read column-wise, gives a second relation that is just as useful in practice:

Theorem 5.6 (Second orthogonality relation). *For any $g, h \in G$,*

$$\sum_i \overline{\chi_i(g)} \chi_i(h) = \begin{cases} \frac{|G|}{|[g]|} = |C_G(g)|, & h \text{ conjugate to } g, \\ 0, & \text{otherwise,} \end{cases}$$

the sum running over all irreducible characters.

Proof. Let $C_1, \dots, C_k$ be the classes and form the $k \times k$ matrix

$$X_{iC} = \sqrt{\frac{|C|}{|G|}} \chi_i(g_C).$$

The first orthogonality relation says precisely that the rows of X are orthonormal, $XX^\dagger = \mathbb{I}$; a square matrix with orthonormal rows is unitary, so its columns are orthonormal too, $X^\dagger X = \mathbb{I}$:

$$\frac{\sqrt{|C| |C'|}}{|G|} \sum_i \overline{\chi_i(g_C)} \chi_i(g_{C'}) = \delta_{CC'},$$

which is the assertion (the centraliser form uses $|[g]| = [G : C_G(g)]$ from Lecture 2). ■

Setting $g = h = e$ recovers $\sum_i n_i^2 = |G|$: the column of the identity carries the dimension theorem of Lecture 4.

5.3 Characters know everything

Theorem 5.7 (Multiplicity formula). *Let (ρ, V) be any representation, with character χ_V and decomposition $V \cong \bigoplus_i m_i V^{(i)}$. Then*

$$m_i = \langle \chi_i, \chi_V \rangle_G :$$

the multiplicities are inner products, computable from the character alone, with no basis-hunting.

Proof. By additivity (Proposition 5.2(iv)), $\chi_V = \sum_j m_j \chi_j$; take the inner product with χ_i and use orthonormality. ■

Corollary 5.8 (The character is a complete invariant). *Two representations of a finite group are equivalent if and only if they have the same character.*

Proof. Equivalent representations have equal characters (Proposition 5.2(i)). Conversely, equal characters give equal multiplicities by Theorem 5.7, and representations with the same multiplicities are both equivalent to the same direct sum. ■

Corollary 5.9 (Irreducibility criterion). $\langle \chi_V, \chi_V \rangle_G = \sum_i m_i^2$. *In particular V is irreducible if and only if $\langle \chi_V, \chi_V \rangle_G = 1$.*

Proof. Expand $\chi_V = \sum m_i \chi_i$ in the orthonormal basis and compute the norm. A sum of squares of nonnegative integers equals 1 precisely when one m_i is 1 and the rest vanish. ■

Example 5.10 (The S_3 permutation representation, in one line). Lecture 3 decomposed the permutation representation of S_3 by an explicit change of basis. Watch characters do it without one. By Lemma 5.4, on the classes $\{e\}$, $\{\text{transpositions}\}$, $\{\text{3-cycles}\}$ (sizes 1, 3, 2) the character is the number of fixed points:

$$\chi_{\text{perm}} = (3, 1, 0).$$

Against the irreducible characters of S_3 (computed in Section 5.4; trivial $(1, 1, 1)$, sign $(1, -1, 1)$, standard $(2, 0, -1)$):

$$\begin{aligned} m_{\text{triv}} &= \frac{1}{6} [1 \cdot 1 \cdot 3 + 3 \cdot 1 \cdot 1 + 2 \cdot 1 \cdot 0] = 1, \\ m_{\text{sgn}} &= \frac{1}{6} [1 \cdot 1 \cdot 3 + 3 \cdot (-1) \cdot 1 + 2 \cdot 1 \cdot 0] = 0, \\ m_{\text{std}} &= \frac{1}{6} [1 \cdot 2 \cdot 3 + 3 \cdot 0 \cdot 1 + 2 \cdot (-1) \cdot 0] = 1, \end{aligned}$$

recovering $\mathbb{C}^3 \cong V^{\text{triv}} \oplus V^{\text{std}}$ — including the *absence* of the sign representation, which the explicit method could only discover after the fact. As a cross-check, $\langle \chi_{\text{perm}}, \chi_{\text{perm}} \rangle_G = \frac{1}{6}(9 + 3 + 0) = 2 = 1^2 + 1^2$: exactly two irreducible constituents.

Example 5.11 (The regular representation, again). With the character (5.1), the multiplicity of $V^{(i)}$ in the regular representation is

$$m_i = \langle \chi_i, \chi_{\text{reg}} \rangle_G = \frac{1}{|G|} \overline{\chi_i(e)} |G| = n_i :$$

the decomposition theorem of Lecture 4, re-derived in one line. This is the promised payoff of building the character machinery: theorems that required structural arguments become arithmetic.

5.4 Character tables

All the character theory of a finite group fits in a single square array: the *character table*, with one row per irreducible character, one column per conjugacy class. Conventionally the first column is the class of the identity — so the first column lists the dimensions n_i — and the first row is the trivial character, all 1's; class sizes are recorded above the columns, since the inner product needs them. The two orthogonality relations say that the table, suitably weighted, is a unitary matrix: rows are orthonormal with weights $|C|/|G|$, columns with weights $|G|/|C|$ — a complete error-check on every entry.

A systematic construction algorithm, with the tools now in hand:

- (1) find the conjugacy classes and their sizes (Lecture 2);
- (2) the number of irreducible representations equals the number of classes; solve $\sum_i n_i^2 = |G|$ with $n_1 = 1$ for the dimension pattern (Lecture 4), which in small groups is usually forced;
- (3) write down the one-dimensional characters: these are homomorphisms $G \rightarrow \mathbb{C} \setminus \{0\}$, fixed by consistent assignments to generators compatible with the relations *and with conjugacy* (a one-dimensional character takes equal values on conjugate elements);
- (4) fill the remaining rows from known subrepresentations (traces of explicit matrices) or by orthogonality to the rows already known — column orthogonality against the identity column is especially effective;
- (5) verify the completed table against both orthogonality relations.

Example 5.12 (C_n , and C_3 in full). For the abelian group C_n every element is its own class, and the n irreducible characters are the one-dimensional representations $\chi_k(r^m) = \omega^{km}$,

$\omega = e^{2\pi i/n}$ (Lectures 3–4). For $n = 3$:

C_3	e	r	r^2
χ_0	1	1	1
χ_1	1	ω	ω^2
χ_2	1	ω^2	ω

Orthogonality of rows is the geometric-series identity $1 + \omega^k + \omega^{2k} = 0$ for $k \neq 0$: character theory on cyclic groups is the arithmetic of roots of unity — the discrete Fourier transform of Lecture 4 in tabular form.

Example 5.13 (S_3). Classes $\{e\}$, $\{(1\ 2), (1\ 3), (2\ 3)\}$, $\{(1\ 2\ 3), (1\ 3\ 2)\}$ of sizes 1, 3, 2; dimension pattern (1, 1, 2) from $1 + 1 + 4 = 6$. The one-dimensional characters are the trivial and the sign. The two-dimensional row can be read off from the standard-representation matrices of Lecture 3, $\text{tr } B((1\ 2)) = 0$ and $\text{tr } B((1\ 2\ 3)) = -1$ — or, knowing no matrices at all, from column orthogonality against the identity column, $\sum_i \overline{\chi_i(e)} \chi_i(c) = 0$ for each class $c \neq \{e\}$: for the transpositions, $1 \cdot 1 + 1 \cdot (-1) + 2 \chi_{\text{std}}((1\ 2)) = 0$ gives $\chi_{\text{std}}((1\ 2)) = 0$; for the 3-cycles, $1 \cdot 1 + 1 \cdot 1 + 2 \chi_{\text{std}}((1\ 2\ 3)) = 0$ gives $\chi_{\text{std}}((1\ 2\ 3)) = -1$. The table:

S_3	1 e	3 (1 2)	2 (1 2 3)
χ_{triv}	1	1	1
χ_{sgn}	1	-1	1
χ_{std}	2	0	-1

(class sizes prefixed to the representatives). Check: $\langle \chi_{\text{std}}, \chi_{\text{std}} \rangle_G = \frac{1}{6}(4 + 0 + 2) = 1$ — irreducible, as proved by hand in Lecture 3.

Example 5.14 (D_4). Classes (Lecture 2): $\{e\}$, $\{r^2\}$, $\{r, r^3\}$, $\{s, r^2s\}$, $\{rs, r^3s\}$, sizes 1, 1, 2, 2, 2; dimension pattern (1, 1, 1, 1, 2), forced (Lecture 4). For the one-dimensional characters ν : conjugacy of r with $r^{-1} = r^3$ forces $\nu(r) = \nu(r)^{-1}$, i.e. $\nu(r) = \pm 1$, and $\nu(s) = \pm 1$ from $s^2 = e$; all four sign choices are consistent with the relations, and $\nu(r^2) = \nu(r)^2 = 1$ always. The two-dimensional row follows from column orthogonality against the identity column: for each class $c \neq \{e\}$,

$$\sum_i \overline{\chi_i(e)} \chi_i(c) = 0 \implies \chi_5(c) = -\frac{1}{2} \sum_{1\text{-dim}} \nu(c),$$

giving $\chi_5(r^2) = -2$ and $\chi_5(r) = \chi_5(s) = \chi_5(rs) = 0$:

D_4	1 e	1 r^2	2 r	2 s	2 rs
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	1	-1	1	-1
χ_4	1	1	-1	-1	1
χ_5	2	-2	0	0	0

As a check, the defining representation $D(r) = R(\pi/2)$, $D(s) = \text{diag}(-1, 1)$ has traces 2, -2, 0, 0, 0 — it is χ_5 , and $\langle \chi_5, \chi_5 \rangle_G = \frac{1}{8}(4 + 4 + 0 + 0 + 0) = 1$ confirms irreducibility. The quaternion group Q_8 , despite being non-isomorphic to D_4 , has an *identical* character table (Sheet 5): the character table is a powerful invariant, but not a complete one for the group itself.

Example 5.15 (C_{3v} , in the chemists' notation). The point group C_{3v} of ammonia is isomorphic to S_3 (Lecture 1), so its character table contains the same numbers; but the applications of Lecture 6 will use the notation in which chemists and spectroscopists tabulate it. Classes are labelled by operation type — the identity E , the two rotations $2C_3$, the three vertical reflections $3\sigma_v$ — and the irreducible representations carry *Mulliken symbols*:

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0

The conventions, in the part we need: A and B denote one-dimensional representations, symmetric ($\chi(C_n) = +1$) respectively antisymmetric (-1) under the principal rotation; the subscript 1 or 2 records symmetry or antisymmetry under the vertical reflections; E denotes a two-dimensional irreducible and T a three-dimensional one. Beware the standard double duty of the letter E : the identity *operation* (a column) and the two-dimensional *representation* (a row); context always disambiguates. Likewise these notes write e for the abstract identity element but E in Schoenflies tables — both usages are entrenched, and we keep each in its habitat.

5.5 The canonical decomposition and projection operators

Characters compute multiplicities; we now make them compute *subspaces*. Two notational conventions, used from here through Lecture 7: superscripts in parentheses, $V^{(i)}$, always denote the abstract irreducible representations; subscripts, V_i , denote the pieces of a given representation V defined next.

Definition 5.16 (Isotypic components). Let (ρ, V) be a representation. The *isotypic component* V_i of type i is the sum of all irreducible invariant subspaces of V isomorphic to $V^{(i)}$.

If $V \cong \bigoplus_i m_i V^{(i)}$, the component V_i collects the m_i copies of $V^{(i)}$, so $\dim V_i = m_i n_i$. The individual copies inside V_i are *not* canonical when $m_i > 1$ (Lecture 3's caveat), but the components themselves are, as the following theorem shows by exhibiting each V_i as the image of an operator built solely from the character.

Theorem 5.17 (Canonical decomposition and projectors). For each irreducible representation define

$$P_i = \frac{n_i}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \rho(g).$$

Then:

- (i) each P_i commutes with every $\rho(h)$ and preserves every invariant subspace;
- (ii) on an irreducible invariant subspace $W \subseteq V$ of type j , P_i acts as $\delta_{ij} \text{id}_W$;
- (iii) $P_i^2 = P_i$, $P_i P_j = 0$ for $i \neq j$, and $\sum_i P_i = \text{id}_V$;
- (iv) P_i is the projection onto the isotypic component V_i along the sum of the others:

$$V = \bigoplus_i V_i, \quad V_i = \text{im } P_i,$$

a decomposition canonically attached to V ;

$$(v) \quad \text{tr } P_i = m_i n_i = \dim V_i.$$

Proof. (i) Conjugating and substituting $g' = hgh^{-1}$,

$$\rho(h)P_i\rho(h)^{-1} = \frac{n_i}{|G|} \sum_g \overline{\chi_i(g)} \rho(hgh^{-1}) = \frac{n_i}{|G|} \sum_{g'} \overline{\chi_i(h^{-1}g'h)} \rho(g') = P_i,$$

because χ_i is a class function. If W is invariant then $\rho(g)W \subseteq W$ for every g , hence $P_i W \subseteq W$.

(ii) By (i), the restriction $P_i|_W$ commutes with the irreducible action of G on W , so Schur's lemma gives $P_i|_W = \lambda \text{id}_W$. The trace determines λ :

$$\lambda n_j = \text{tr}(P_i|_W) = \frac{n_i}{|G|} \sum_g \overline{\chi_i(g)} \chi_j(g) = n_i \langle \chi_i, \chi_j \rangle_G = n_i \delta_{ij},$$

so $\lambda = \delta_{ij}$.

(iii)–(iv) Decompose V into irreducible invariant subspaces (Maschke). By (ii), P_i restricts to the identity on each type- i summand and to zero on every other summand; all the algebraic relations in (iii) follow by checking them summand by summand, and the image of P_i is exactly the span of the type- i summands. That span contains every irreducible subspace $U \cong V^{(i)}$ of V whatsoever (again by (ii), $P_i|_U = \text{id}$ gives $U \subseteq \text{im } P_i$), so $\text{im } P_i$ is the isotypic component of Definition 5.16 — and the formula for P_i makes no reference to the chosen decomposition, so V_i is canonical.

$$(v) \quad \text{tr } P_i = \frac{n_i}{|G|} \sum_g \overline{\chi_i(g)} \chi_V(g) = n_i \langle \chi_i, \chi_V \rangle_G = n_i m_i \text{ by the multiplicity formula.} \quad \blacksquare$$

The identity $\sum_i P_i = \text{id}_V$ has a pleasant direct verification: by the second orthogonality relation (column of the identity), $\sum_i n_i \overline{\chi_i(g)} = |G| \delta_{g,e}$, so the sum collapses to $\rho(e)$.

Example 5.18 (The three projectors of the S_3 permutation representation). Take $V = \mathbb{C}^3$ with the permutation matrices of Lecture 3, and let J denote the 3×3 matrix of ones. Summing the six permutation matrices puts a 2 in every entry (for fixed (i, j) , exactly two permutations send j to i), so

$$P_{\text{triv}} = \frac{1}{6} \sum_g \rho(g) = \frac{1}{3} J,$$

the rank-one projection onto the diagonal $\mathbb{C}(1, 1, 1)$ — precisely the “average over the group” operator. For the standard component, with $C = D((1\ 2\ 3))$ and using $\mathbb{K} + C + C^2 = J$ (the three cyclic matrices have disjoint supports filling J):

$$P_{\text{std}} = \frac{2}{6} [2\mathbb{K} - C - C^2] = \frac{1}{3} (3\mathbb{K} - J) = \mathbb{K} - \frac{1}{3} J,$$

the projection onto the plane $\sum_i v_i = 0$. And the three transposition matrices likewise sum to J (each entry of J hit exactly once), so

$$P_{\text{sgn}} = \frac{1}{6} [\mathbb{K} + C + C^2 - (\text{sum of transposition matrices})] = \frac{1}{6} (J - J) = 0 :$$

the projector detects the absence of the sign representation by vanishing identically. The checks of Theorem 5.17 are immediate: $P_{\text{triv}} + P_{\text{std}} + P_{\text{sgn}} = \mathbb{K}$, the traces are $1, 2, 0 = m_i n_i$, and each operator squares to itself.

5.5.1 Finer projectors: partner functions

The character projector P_i lands in the isotypic component but does not see inside it. When the matrices $D^{(i)}$ themselves are known (not just their traces), one can resolve V_i direction by direction. Fix unitary matrices $D^{(i)}$ and define, for $1 \leq \mu, \nu \leq n_i$,

$$P_{\mu\nu}^{(i)} = \frac{n_i}{|G|} \sum_{g \in G} \overline{D_{\mu\nu}^{(i)}(g)} \rho(g).$$

Call a basis $(w_1, \dots, w_{n_i})$ of an irreducible subspace $W \cong V^{(i)}$ *adapted* (its members *partner functions*) if the action takes the standard matrix form $\rho(g)w_\nu = \sum_\mu D_{\mu\nu}^{(i)}(g) w_\mu$.

Theorem 5.19 (Matrix-element projectors). (i) $P_{\mu\nu}^{(i)} P_{\alpha\beta}^{(j)} = \delta_{ij} \delta_{\nu\alpha} P_{\mu\beta}^{(i)}$; in particular the diagonal operators $P_{\mu\mu}^{(i)}$ are commuting projections, mutually orthogonal across all (i, μ) ;

(ii) $\sum_{\mu=1}^{n_i} P_{\mu\mu}^{(i)} = P_i$;

(iii) on an adapted basis of any irreducible subspace of type i ,

$$P_{\mu\nu}^{(i)} w_\beta = \delta_{\nu\beta} w_\mu,$$

while every $P_{\mu\nu}^{(i)}$ annihilates irreducible subspaces of type $j \neq i$. Thus $P_{\mu\mu}^{(i)}$ projects onto the “ μ -th direction” of each type- i copy, and $P_{\mu\nu}^{(i)}$ transfers the ν -th partner into the μ -th: a ladder between partner functions.

Proof. (iii) first, since it is the cleanest:

$$P_{\mu\nu}^{(i)} w_\beta = \frac{n_i}{|G|} \sum_g \overline{D_{\mu\nu}^{(i)}(g)} \sum_\alpha D_{\alpha\beta}^{(j)}(g) w_\alpha = \sum_\alpha \left[n_i \left\langle D_{\mu\nu}^{(i)}, D_{\alpha\beta}^{(j)} \right\rangle_G \right] w_\alpha = \delta_{ij} \delta_{\nu\beta} w_\mu$$

by the great orthogonality theorem. (ii) follows by summing the definition over $\mu = \nu$, since $\sum_\mu D_{\mu\mu}^{(i)} = \chi_i$. For (i), evaluate both sides on an adapted basis of each irreducible summand of V (Maschke): by (iii), the left side sends w_β (type k) to $\delta_{jk} \delta_{\alpha\beta} P_{\mu\nu}^{(i)} w_\alpha = \delta_{jk} \delta_{\alpha\beta} \delta_{ik} \delta_{\nu\alpha} w_\mu$, and the right side to $\delta_{ij} \delta_{\nu\alpha} \delta_{ik} \delta_{\alpha\beta} w_\mu$ — equal. Two operators agreeing on a basis of V are equal. ■

Unlike P_i , the finer projectors depend on the chosen matrix realisation $D^{(i)}$ — rotate the basis of $V^{(i)}$ and the $P_{\mu\nu}^{(i)}$ rotate accordingly; only their sum P_i is canonical. In applications this is a feature, not a bug: one chooses the $D^{(i)}$ adapted to the physics (say, aligned with molecular axes), and then uses enough independent starting vectors to span every multiplicity space. Fix a reference index ν for type i , apply $P_{\nu\nu}^{(i)}$ to candidate vectors, discard zero results, and retain (or orthonormalise) independent results until they span $\text{im } P_{\nu\nu}^{(i)}$. For each retained vector u , the transfer operators $P_{\mu\nu}^{(i)}$ manufacture its partners $P_{\mu\nu}^{(i)} u$ for all μ . Repeat for every type i ; the resulting vectors span the whole representation. A single seed can miss an isotypic component or fail to resolve repeated copies, whereas this rank-aware procedure cannot. Lecture 6 practises the coarser, canonical character/isotypic-projector method on the normal modes of H_2O and NH_3 ; it does not carry out this finer rank-aware construction. Lecture 7 then uses the resulting isotypic organisation for degenerate quantum levels.

Remark 5.20 (Computer algebra: the whole lecture as a notebook). Everything above mechanises. The accompanying Mathematica session (i) recomputes conjugacy classes by the

brute-force gather of Lecture 2; (ii) assembles candidate character tables and verifies both orthogonality relations as matrix identities, $XX^\dagger = X^\dagger X = \mathbb{I}$ for the weighted table X ; (iii) computes characters of given representations as traces, multiplicities as the inner products of Theorem 5.7, and irreducibility via $\langle \chi, \chi \rangle_G$; and (iv) builds the projectors P_i and $P_{\mu\nu}^{(i)}$ as explicit matrices, confirming Example 5.18 — $P_{\text{triv}} = J/3$, $P_{\text{std}} = \mathbb{I} - J/3$, $P_{\text{sgn}} = 0$ — together with the rank and idempotency checks. Exact arithmetic with root-of-unity entries (`Exp[2 Pi I k/n]`, `FullSimplify`) keeps every verification symbolic.

Summary.

- *Character.* $\chi(g) = \text{tr } \rho(g)$, a class function invariant under equivalence (Definition 5.1, Proposition 5.2).
- *Orthogonality.* Irreducible characters are orthonormal (first relation, Theorem 5.5); their columns obey a dual relation (Theorem 5.6).
- *Multiplicities.* $\langle \chi, \chi_i \rangle$ is the multiplicity of the i th irreducible (Theorem 5.7); $\|\chi\|^2 = 1$ iff the representation is irreducible (Corollary 5.9).
- *Complete invariant.* The character determines a representation up to equivalence (Corollary 5.8).
- *Projectors.* Character projectors give the canonical decomposition into isotypic components (Theorem 5.17).

Tutorial. Exercise Sheet 5.

References

Serre, *Linear Representations of Finite Groups*, Pt. I (character theory); Jones Pt. 5; Tung Ch. 3–4.

Bibliography

The per-lecture *References* sections cite the works below by author and short title; full details are collected here. Chapter and section numbers in the lecture references follow these editions.

- M. A. Armstrong, *Groups and Symmetry*. Springer (UTM), 1988.
- V. Bargmann, On unitary ray representations of continuous groups. *Annals of Mathematics* **59** (1954), 1–46.
- A. O. Barut and R. Rączka, *Theory of Group Representations and Applications*, 2nd ed. World Scientific, 1986.
- R. Berndt, *Representations of Linear Groups: An Introduction Based on Examples from Physics and Number Theory*. Vieweg, 2007. DOI: 10.1007/978-3-8348-9401-4. Theorem 5.2, p. 71 gives the SNAG theorem used for the translation subgroup.
- D. M. Bishop, *Group Theory and Chemistry*. Dover, 1993 (orig. Oxford University Press, 1973).
- O. Bratteli and D. W. Robinson, *Operator Algebras and Quantum Statistical Mechanics 2: Equilibrium States, Models in Quantum Statistical Mechanics*, 2nd ed. (1997), 2nd printing. Springer, 2002. Section 5.2.1, pp. 7–8 gives Fock-space second quantisation, its finite-particle analytic core and closure, and the lifted unitary group.
- G. Costa and G. Fogli, *Symmetries and Group Theory in Particle Physics*. Springer (Lecture Notes in Physics 823), 2012.
- J. B. Conway, *A Course in Functional Analysis*, 1st ed. Springer (GTM 96), 1985. Chapter IX, Theorems 6.4 and 6.6 give the strong/weak bicommutant closure and the multiplication maximal-abelian theorem for sigma-finite measure spaces.
- J. P. Elliott and P. G. Dawber, *Symmetry in Physics*, 2 vols. Macmillan, 1979.
- K. Erdmann and M. J. Wildon, *Introduction to Lie Algebras*. Springer (SUMS), 2006.
- W. Fulton and J. Harris, *Representation Theory: A First Course*. Springer (GTM 129), 1991.
- G. B. Folland, *Harmonic Analysis in Phase Space*. Princeton University Press (Annals of Mathematics Studies 122), 1989. Chapter 1, Sections 2–5, including Theorem 1.50, treats the Heisenberg group, Schrödinger/Weyl representation, and Stone–von Neumann theorem.
- G. B. Folland, *A Course in Abstract Harmonic Analysis*, 2nd ed. Chapman & Hall/CRC, 2016. Section 6.6, Theorem 6.42, pp. 199–201 gives the regular semidirect-product Mackey theorem, including induced irreducibility, exhaustivity, and equivalence.
- R. Gilmore, *Lie Groups, Lie Algebras, and Some of Their Applications*. Wiley, 1974 (repr. Dover, 2005).
- B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed. Springer (GTM 222), 2015.
- N. Jeevanjee, *An Introduction to Tensors and Group Theory for Physicists*, 2nd ed. Birkhäuser, 2015.
- H. F. Jones, *Groups, Representations and Physics*, 2nd ed. Institute of Physics Publishing,

1998. Chapter 8, Sections 8.3–8.4, pp. 149–164 is the course’s explicit $SU(3)$ further-reading route; Chapter 10, Section 10.4, pp. 208–216 treats the Poincaré little groups; Appendix E, pp. 279–280 derives the covariant on-shell measure from $\theta(p^0)\delta(p^2 - m^2)d^4p$.
- Z.-Q. Ma, *Group Theory for Physicists*. World Scientific, 2007.
 - A. M. L. Messiah and O. W. Greenberg, Symmetrization Postulate and Its Experimental Foundation. *Physical Review* **136** (1964), B248–B267. DOI: <https://doi.org/10.1103/PhysRev.136.B248>.
 - H. Osborn, *Group Theory*. University of Cambridge, Mathematical Tripos Part III lecture notes (available from the DAMTP lecture-notes pages).
 - M. Reed and B. Simon, *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, rev. ed. Academic Press, 1980. Chapter VIII, Theorems VIII.5, VIII.8, VIII.10, and VIII.13 and Section VIII.5, Example 2 cover spectral calculus, Stone’s theorem, invariant-domain generators, strong commutation, and the bounded-CCR obstruction.
 - P. Ramond, *Group Theory: A Physicist’s Survey*. Cambridge University Press, 2010. Chapter 11, Section 11.1.1, pp. 224–231 treats the Pauli–Lubanski Casimirs and the massless little group; pp. 230–231 distinguish finite helicity from continuous spin and record the two integer/half-odd-integer cover sectors.
 - H. Rauch, A. Zeilinger, G. Badurek, A. Wilfing, W. Bauspiess, and U. Bonse, Verification of Coherent Spinor Rotation of Fermions. *Physics Letters A* **54** (1975), 425–427. DOI: 10.1016/0375-9601(75)90798-7.
 - J. Schwichtenberg, *Physics from Symmetry*, 2nd ed. Springer, 2018.
 - J.-P. Serre, *Linear Representations of Finite Groups*. Springer (GTM 42), 1977.
 - J. Stillwell, *Naive Lie Theory*. Springer (UTM), 2008.
 - E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*. Princeton University Press (Princeton Lectures in Analysis I), 2003. Chapter 3, Section 1, Theorem 1.1, pp. 69–79 gives the mean-square completeness result used in Lecture 12.
 - M. Tinkham, *Group Theory and Quantum Mechanics*. McGraw–Hill, 1964 (repr. Dover, 2003).
 - W.-K. Tung, *Group Theory in Physics*. World Scientific, 1985. Chapter 10, Section 10.4, pp. 191–200 treats induced Poincaré representations, Pauli–Lubanski brackets, and massive and massless little groups.
 - S. Weinberg, *The Quantum Theory of Fields, Vol. I: Foundations*. Cambridge University Press, 1995. Section 5.6 treats causal fields and the relativistic spin–statistics connection, a theorem not derived from a rotation phase in this course.
 - S. A. Werner, R. Colella, A. W. Overhauser, and C. F. Eagen, Observation of the Phase Shift of a Neutron Due to Precession in a Magnetic Field. *Physical Review Letters* **35** (1975), 1053–1055. DOI: 10.1103/PhysRevLett.35.1053.
 - P. Woit, *Quantum Theory, Groups and Representations: An Introduction*. Springer, 2017.